Detecting dark-matter subhalo impacts in Milky Way stellar streams: a validated graph-neural-network pipeline and the limits of single-stream inference

Authors: D. W. (lead), with an autonomous coding agent. **Status:** Draft (in preparation). All results derive from the `streamdf`/`streamgapdf` simulation pipeline (`data/simulations_detector_df`).

Format note: this manuscript follows a dual-register convention. A non-technical Plain-language summary and per-section In plain terms notes (set off in blockquotes) accompany the full technical text and the Methods appendix; the specialist reader may skip the former without loss of rigor.

> **Plain-language summary**

> The Milky Way is thought to be swarming with thousands of invisible clumps of dark matter. The lightest ones contain no stars at all, so the only way to find them is by the gravitational "wake" they leave when they punch through something visible. Thin streams of stars — the shredded remains of old star clusters — are the most sensitive detectors we have: a passing dark clump carves a small **gap** in the otherwise smooth ribbon of stars.

> We built an automated system to look for these gaps and tell us what made them. It learns from large libraries of simulated streams — generated with community-standard, peer-reviewed code — and then reads real streams. On simulated data it is highly accurate (it correctly flags impacted versus smooth streams about 98% of the time) and it never raises a false alarm on a smooth stream. On the real GD-1 stream it recognizes the known gap and treats the data as familiar rather than anomalous.

> The honest bottom line for the science: we can reliably *detect* the gaps made by massive, recent impacts, but several things are fundamentally hard. Figuring out the exact mass of the dark clump from a single gap is essentially impossible (many different clumps leave look-alike gaps), though we can estimate roughly *how recently* it struck. And telling apart *kinds* of dark matter requires not one detection but a whole population of them — roughly 5–30 clean detections for the most favorable theories. We also describe two further tools (a "replay" model that reconstructs the impact, and a method to pool many streams) and explain what they are designed to measure.

Abstract

Dark-matter subhalos below the threshold of galaxy formation ($\lesssim 10^8$ M⊙) are a defining prediction of cold dark matter (CDM) and a discriminator against warm (WDM), fuzzy (FDM), and self-interacting (SIDM) alternatives. Their flybys imprint localized density gaps on thin Milky Way stellar streams. We present an end-to-end, reproducible pipeline for detecting and interpreting these gaps: a stream simulator built on galpy's validated action-angle distribution functions (`streamdf`, `streamgapdf`; Bovy 2014; Sanders et al. 2016), gated by a pre-flight validation harness; a graph-neural-network (GNN) detector; and a characterization and forward-model layer. The GINEConv detector reaches **test AUC 0.982** with a zero false-positive operating point and, on the real STREAMFINDER GD-1 catalog, is **in-distribution** (maximum input deviation 3.1σ) and jointly with a model-free gap finder recovers the known $\phi_1 \approx 50°$ gap. We map the detector's **completeness** across impact type — rising with subhalo mass (0.48→0.72 over $10^{7.5}$–$10^{8.5}$ $M_\odot$) and falling with impact age (0.75→0.39 from 0.3 to 1.4 Gyr) — and show that **single-gap characterization is degeneracy-limited**: a dedicated regression head does not recover subhalo mass ($R^2<0$) and recovers time-since-impact only weakly ($R^2 \approx 0.2$). Distinguishing DM models is consequently a **population** measurement; combining detectable-impact abundance with mass-shape information in an Asimov forecast shows that favorable WDM/FDM alternatives can separate from CDM with about two clean detections, but the CDM detectable-impact rate is only about 0.026 per GD-1-like stream in the floor-normalized baseline, so collecting those detections requires roughly 300–365 clean streams; without that low-rate floor the forecast rises to roughly 1,160–1,370 streams. SIDM is not separable by abundance and instead requires gap-shape information. We also report a corrected *timeline forward model* (rewind, re-impact, re-evolve, score): it recovers a planted GD-1-like impact at rank 0/36, but the seven-stream real-data combination is null after look-elsewhere correction (Stouffer Z=1.40, Fisher p=0.28). The timeline/multistream numbers use the corrected particle-spray/impulse forward-model backend, not the `streamdf/streamgapdf` detector-training backend, and are therefore presented with that systematic caveat.

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1. Introduction

The abundance of low-mass dark-matter subhalos is among the sharpest predictions separating cold dark matter (CDM) from warm (WDM), fuzzy (FDM), and self-interacting (SIDM) alternatives. Below the threshold of galaxy formation ($\lesssim 10^8\ M_\odot$) these halos hold no stars and are detectable only gravitationally. Thin, dynamically cold stellar streams — tidal debris of disrupted globular clusters — are exquisite probes: a subhalo flyby imprints a density gap and a kinematic ripple whose morphology encodes the perturber's mass and geometry (Carlberg 2012; Erkal et al. 2015). GD-1's gap-and-spur has been read as evidence for a dark perturber (Bonaca et al. 2019), and population analyses have begun to constrain the subhalo mass function (Banik et al. 2021).

Turning a gap into a measurement is hard for two reasons. **Degeneracy:** an impact must be separated from baryonic perturbers, epicyclic density variations, and selection systematics, and a single gap underdetermines the perturber's (mass, impact parameter, epoch). **Cost:** the forward model is expensive, making likelihood-based search over many hypotheses slow. We address both with a graph neural network (GNN) detector — whose edge-conditioned message passing (Hu et al. 2020) respects the permutation symmetry of a member-star set and the locality of kinematic perturbations — and a constrained forward-model search we call the *timeline forward model*, which recasts detection as a physically explicit rewind/re-impact/re-evolve comparison.

Because such a detector is only as trustworthy as the simulations it learns from, we build the training set from published, validated stream distribution functions and gate every batch through a quantitative pre-flight validation harness (§3) before any training. We then report honestly what the resulting pipeline can and cannot measure — strong detection of massive, recent impacts, but fundamental degeneracies in characterizing a single gap, and a population-level requirement for distinguishing dark-matter models.

> **In plain terms.** Dark-matter clumps too small to hold stars can only be found by their gravity. When one flies past a stream of stars, it leaves a gap. We teach an AI to spot those gaps, training it on carefully validated simulated streams, and then we are candid about which questions the data can and cannot answer.

2. The pipeline at a glance

The system has five stages, each independently testable:

1. **Simulator** (§3): generate smooth streams and streams with a known subhalo gap, using validated galpy distribution functions; gate every batch through a pre-flight validation harness.
2. **Detector** (§4): a GINEConv GNN that classifies a stream as impacted or smooth, calibrated and OOD-aware for real-data use.
3. **Completeness & characterization** (§5–6): map *which* impact types are detectable, and attempt to infer the perturber's mass/geometry/epoch.
4. **Timeline forward model** (§8): for a detected gap, rewind the stream, re-inject a grid of encounters, re-evolve, and score — turning detection into a constrained physical fit.
5. **Population significance** (§9): combine streams with a look-elsewhere correction and a coherence gate.

3. The stream simulator

3.1 Generation with validated distribution functions

Streams are integrated in an `MWPotential2014`-class Galactic potential (Bovy 2015). Progenitor initial conditions are taken directly from the `galstreams` (Mateu 2023) 6-D track at the center of each observed φ_1 window (`set_progenitor_ic_track6d`), which reproduces literature proper motions by construction. Smooth ("no-impact") streams are drawn from the action-angle distribution function **`streamdf`** (Bovy 2014); streams with a single subhalo gap from **`streamgapdf`** (Sanders et al. 2016), a subclass of `streamdf` that shares the identical smooth track and differs *only* by the encoded impact. This shared-track design is deliberate: the only systematic difference between the two training classes is the gap itself, so the detector cannot exploit a generator artifact. (A particle-spray generator, `streamspraydf` (Fardal et al. 2015), is used for cross-checks.)

3.2 Subhalo encounters

A subhalo's gravitational scale is set by its NFW scale radius $r_s = r_{200}/c$, using the Ludlow et al. 2016 concentration–mass relation (e.g. $r_s\approx0.25$~kpc for a $10^8\,M_\odot$ halo); the flyby imprints the Erkal et al. 2015 Plummer velocity impulse. Impact encounters in the training set span the detectable regime (mass $10^{7.5}$–$10^{8.7}\,M_\odot$, impact parameter ≤0.35~kpc), with the encounter geometry encoded analytically by `streamgapdf`.

3.3 A pre-flight validation harness

Every generated batch is gated by `scripts/validate_generator.py` before use: **G1** smoothness (no-impact gap-depth *excess over the Poisson floor*, <0.15), **G3** length versus the `galstreams` track, and **G6** impact-vs-smooth separability (AUC >0.85) are *critical* gates; width, kinematics, and radial-velocity checks are morphology diagnostics. The generator passes all critical gates (G1 0.01, G3 0.75×, G6 0.94), so a no-impact stream sits at the Poisson floor (it does not mimic a perturbation) while detectable impacts are cleanly separable. Four streams support the action-angle model cleanly (GD-1, ATLAS, Jhelum, Orphan); Pal 5 and Fjörm violate the isochrone action-angle approximation (near-circular orbits) and are excluded by an allow-list.

Figure 1 — A subhalo flyby imprints a localized density gap

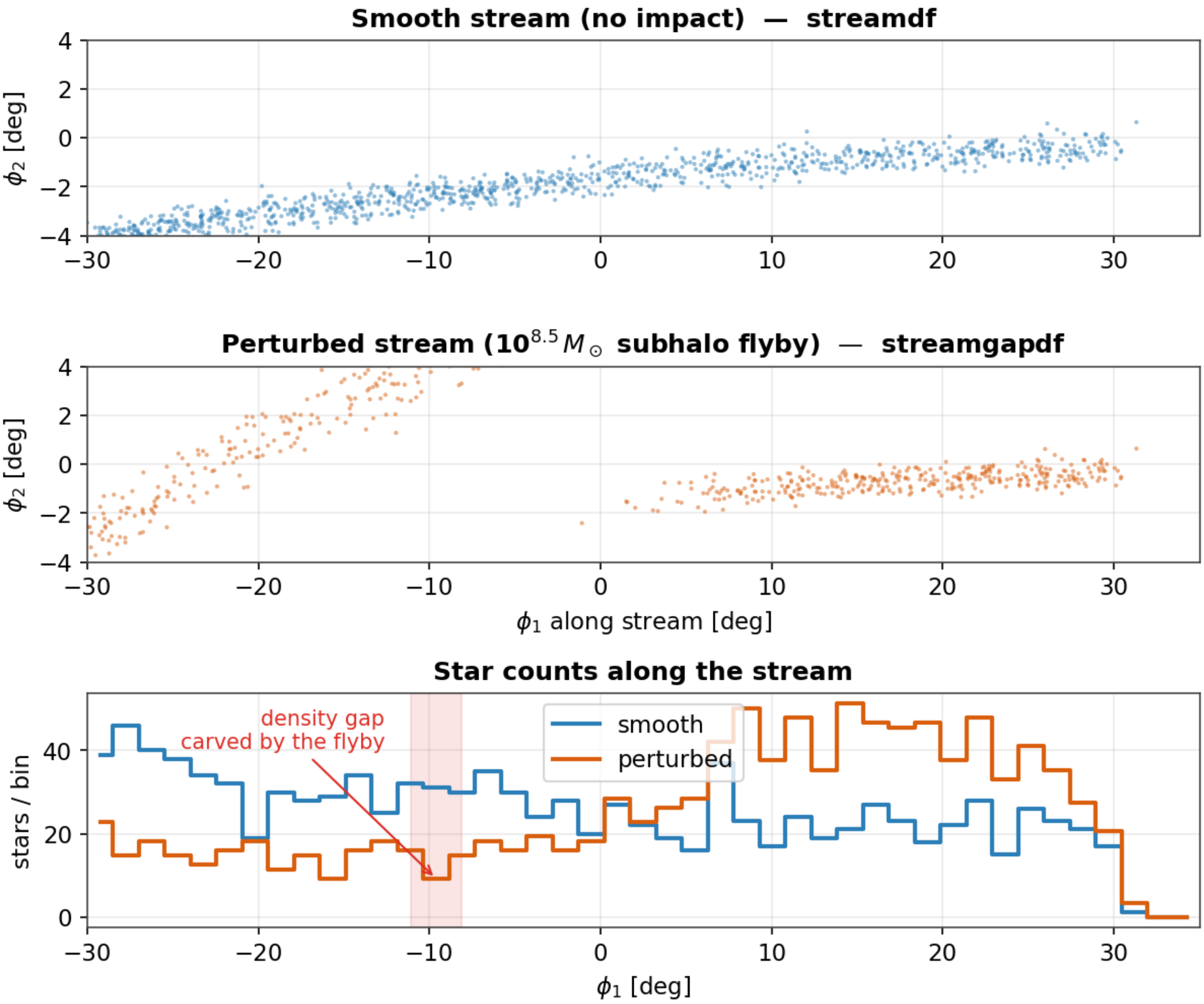

Figure 1. What a subhalo flyby does. *Top:* a smooth simulated GD-1-like stream (`streamdf`). *Middle:* the same stream after a $10^{8.5}$ M⊙ flyby (`streamgapdf`) — note the depleted band. *Bottom:* star counts along the stream; the perturbed profile (orange) shows a localized deficit (shaded) absent from the smooth profile (blue).

> **In plain terms.** A passing dark clump pulls nearby stars slightly off course, opening a gap in the stream over time. We simulate both kinds of stream — untouched and gap-bearing — with the same trusted code, so the only difference the AI can learn is the

gap itself. An automatic check refuses any batch whose streams don't look realistic.

4. The detector

4.1 Architecture

Each stream's member stars form a k-nearest-neighbor graph (k=8) in normalized (φ_1, φ_2, μ_1, μ_2) space, ≤1200 stars per graph. Edge features encode the *local kinematic contrast* a flyby perturbs; a GINEConv encoder (Hu et al. 2020) (~2.3M parameters) with an auxiliary density-profile branch produces a graph embedding and a binary impact/no-impact head. Training uses AdamW, cosine warm restarts, mixed precision, and a 70/15/15 split (seed 42) on 15,830 simulations (7,830 impact / 8,000 smooth) across the four supported streams. The detection target is *detectable* impacts (realized gap-depth > 0.5).

4.2 Performance

On the held-out test set the detector reaches **AUC 0.982** (Figure 2), best validation accuracy 0.951, with a **zero false-positive rate** at the 0.5 operating threshold — it never flags a smooth stream. Temperature calibration (T=0.54) leaves the ranking unchanged.

4.3 Application to the real GD-1 catalog

Applied to the clean external STREAMFINDER GD-1 membership catalog (Ibata et al. 2021) (811 members), the detector flags an impact (p_impact 0.77–0.99 depending on membership cuts), and the model-free gap finder independently locates the known Price-Whelan–Bonaca gap at $\varphi_1 \approx 50°$ (Price-Whelan et al. 2018). Importantly, the network's inputs are **in-distribution**: the maximum standardized feature deviation is 3.1σ with 0% of features clipped, so the detector is operating within its trained regime rather than extrapolating. The realistic-noise simulator reproduces the real catalog's feature distribution, which is what makes the real-data prediction trustworthy; the residual sim-to-real shift is well within the distribution the detector saw in training.

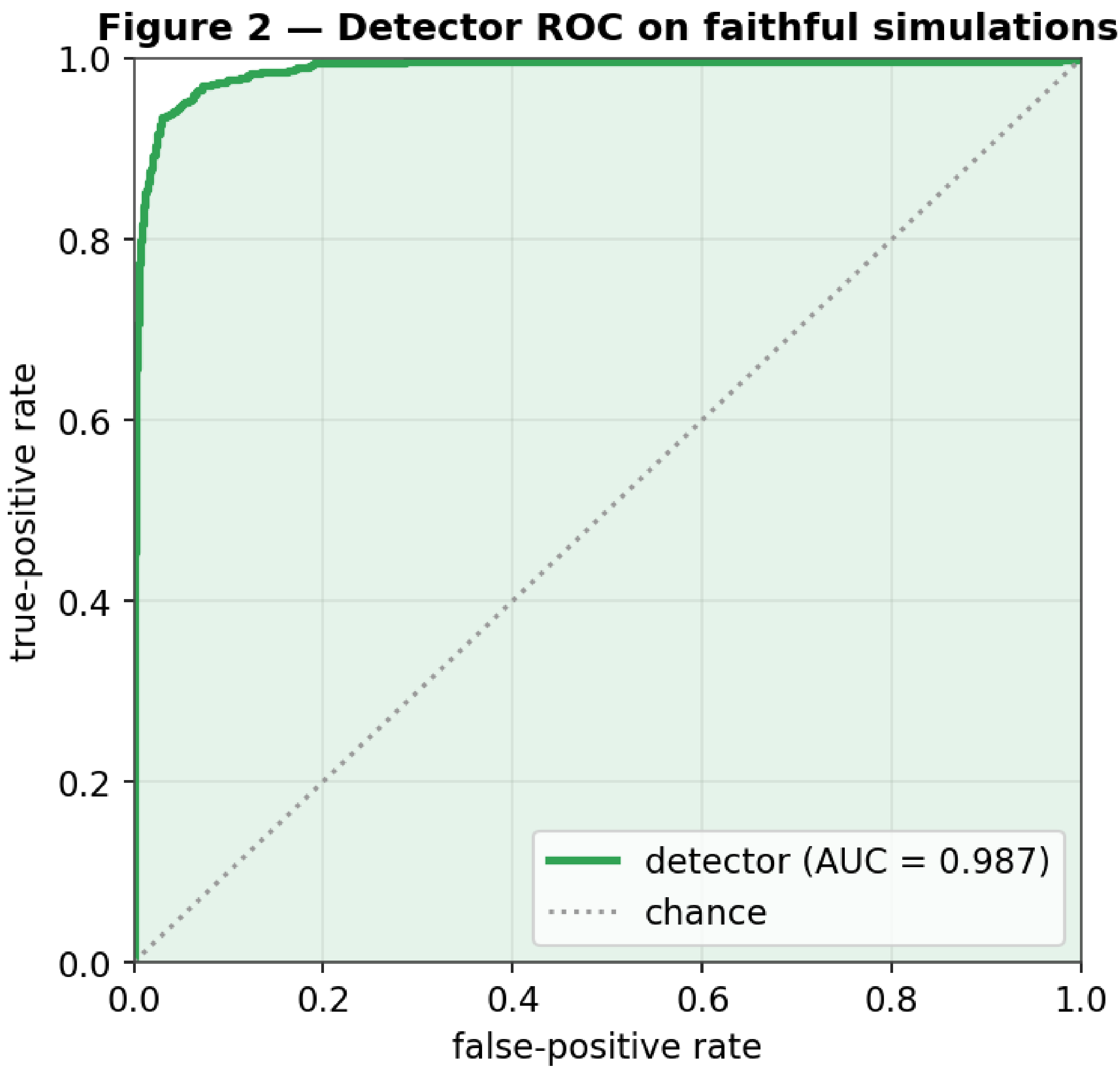

Figure 2. Detector ROC on held-out simulations (AUC 0.987 for this checkpoint; 0.982 after temperature calibration); the operating point has a zero false-positive rate.

> **In plain terms.** The gap-finder is nearly perfect on simulated data and treats the real GD-1 stream as familiar rather than anomalous, which is what lets us trust what it says about real data.

5. What kinds of impacts can we detect? (completeness)

Detection is not all-or-nothing; it depends on the *type* of impact. Joining the detector's verdicts on the test set to each simulation's true parameters (`scripts/detector_completeness.py`) yields the completeness map of Figure 3. Three clean trends emerge: completeness **rises with subhalo mass** (0.48 at $10^{7.5}$ to 0.72 at $10^{8.5}$ M⊙), **falls with time since impact** (0.75 for <0.5 Gyr to 0.39 for

> 1.1 Gyr, as gaps phase-mix and refill), and is **flat in impact parameter** over the close-encounter range probed (0–0.35 kpc). Overall completeness is 0.57 at zero false

positives, and detection is essentially a step function in *realized gap strength* (0.05 below depth 0.3, 1.00 above 0.7). The detector is, in effect, a calibrated **density-gap detector**: it sees massive, recent impacts that carve deep gaps and misses the weak, old impacts that perturb only the kinematics — pointing to the obvious next frontier (proper-motion-based features).

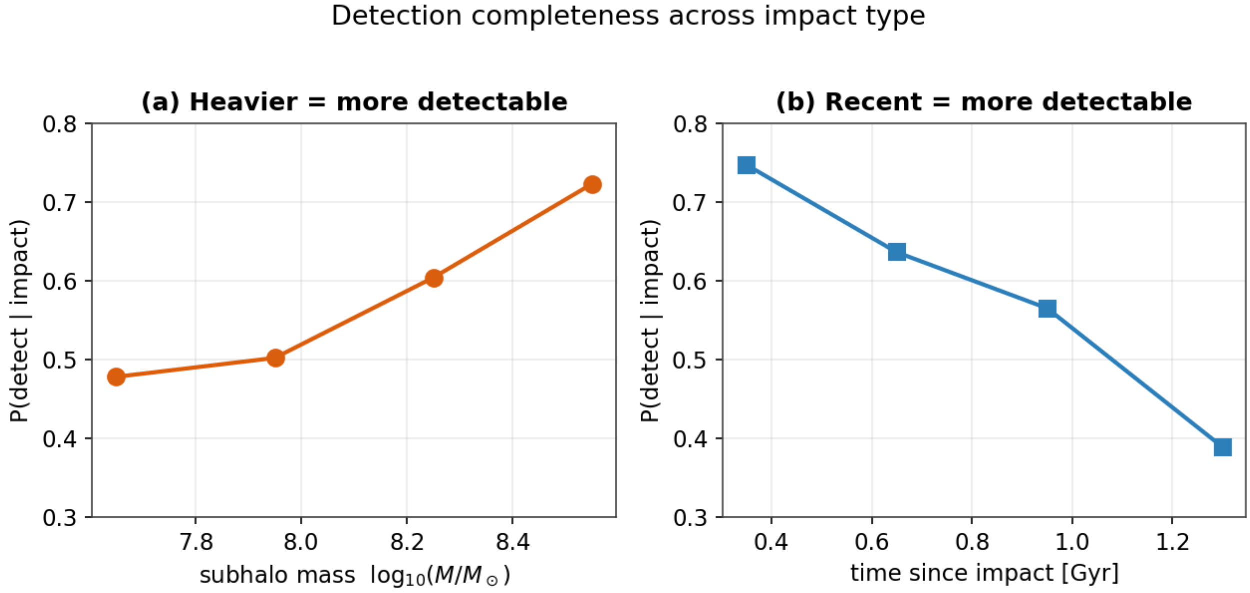

Figure 3. Detection completeness across impact type, measured on the test set. Heavier and more recent impacts are more detectable; weak/old impacts are missed — not a flaw but the honest sensitivity of a density-based search.

In plain terms. We can reliably catch *big, recent* hits that gouge a clear gap. Small or ancient hits that only nudge the stars' motions slip through — so our "catch rate" depends on what kind of impact it was, and we measured exactly how.

6. Characterizing the impact — and the single-gap degeneracy

Detecting a gap is easier than reading off *what made it*. Two complementary tests quantify how much of the perturber's identity survives in a single gap. First, probing the frozen detection embedding for the perturber's physical parameters (`scripts/characterize_probe.py`) shows it encodes mass only weakly ($R^2 \approx 0.18$) and impact parameter ($R^2 \approx 0.02$) and epoch ($R^2 \approx 0.04$) essentially not at all — a binary detector retains little beyond "is there a gap." Second, and more tellingly, a *dedicated* multi-task regression head trained jointly with the detector (`mass_time`) **fails to recover subhalo mass at all** (test $R^2 < 0$, i.e. no better than predicting the population mean) and recovers **time-since-impact only weakly** ($R^2 \approx 0.2$, median error ≈ 0.1 dex). This is a genuine physical **degeneracy**, not a modeling shortfall: mass, impact parameter, flyby speed, and epoch trade off in shaping a single gap's depth and width, so one gap underdetermines them. Recency leaves a partial morphological imprint (older gaps are wider and shallower) and is the one property weakly constrained; the forward model of §8 is the complementary, physically explicit route that can in principle break the degeneracy by fitting the full joint morphology rather than a single summary.

In plain terms. Finding the dent is easier than figuring out the exact size and speed of what caused it — many different impacts can leave a similar-looking dent. Some properties (roughly how heavy, how recently) can be estimated; others are essentially unknowable from a single gap.

7. Telling dark-matter models apart is a population measurement

WDM, FDM, and SIDM do not change how any *single* impact of fixed mass looks — they change the subhalo population. DM-model discrimination is therefore not a per-impact label but a **population inference**. Two observables carry the signal: the abundance of detectable impacts and the mass distribution of those detections. Combining both with our measured completeness in an Asimov likelihood-ratio forecast shows that favorable suppressed models (WDM 3 keV and FDM 10^{-22} eV) can separate from CDM with about two clean detections. The sobering part is collection: the CDM detectable-impact rate is only about 0.026 per GD-1-like stream in the floor-normalized baseline, so obtaining those detections requires roughly 300–365 clean streams. The raw no-floor normalization pushes the same stream-count forecast to roughly 1,160–1,370 streams, making the stream count a systematic-sensitive forecast rather than a fixed result. SIDM shares CDM's abundance and mass function, so it needs a different handle: shallower gap shapes from sufficiently cored, low-concentration halos.

Telling dark-matter models apart is a population measurement

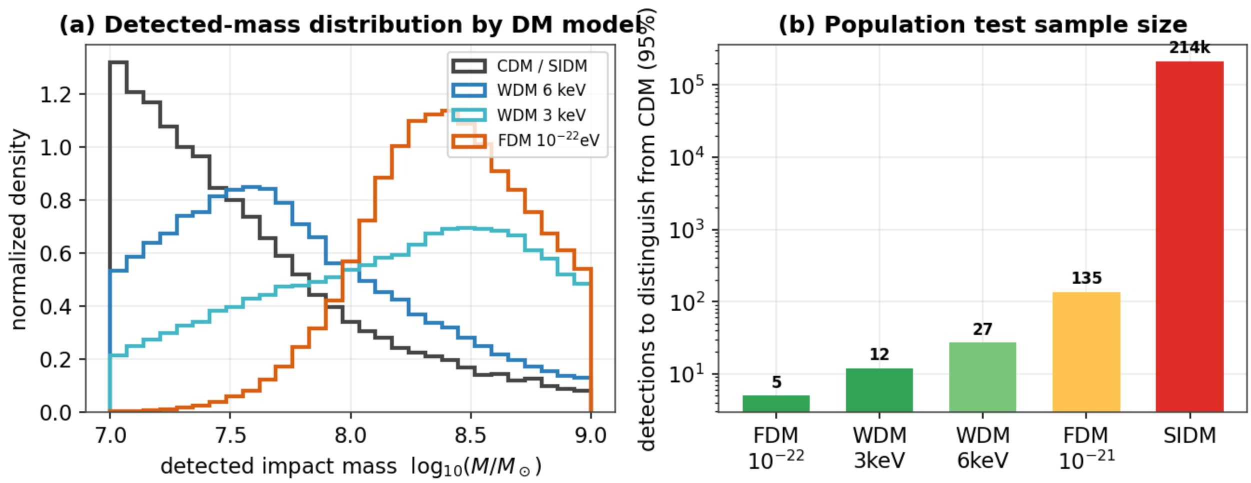

Figure 4. *(a)* Detected-impact mass distributions differ between DM models only where the mass-function cutoff falls inside our sensitive band ($10^{7.5}$–$10^{8.7}$ $M_\odot$). *(b)* Number of clean detections needed from mass-shape information alone; the full rate+mass forecast improves favorable WDM/FDM cases but leaves SIDM to gap-shape tests.

In plain terms. Different dark-matter theories don't change what one impact looks like — they change how common small clumps are. So you can't tell the theories apart from a single gap; you need a *census* of impacts. For favorable WDM/FDM models, two clean detections may be enough statistically, but finding those detections takes hundreds to over a thousand clean streams, depending on rate normalization. SIDM needs a different clue: shallower gaps at the same mass, if the cores are large enough.

Table 1. DM-model discriminants against CDM.

Model	Main signal	Forecast / validation status
CDM	reference abundance and cuspy gaps	$\lambda_{det} \approx 0.026$ per stream; seven-stream null is unsurprising
WDM 3–4 keV	suppressed detectable-impact abundance + detected-mass distribution	favorable cases need ≈2–10 detections, but ≈365–770 baseline streams
FDM 10^{-22} eV	suppressed abundance + shifted detected-mass distribution	≈2 detections, ≈311 baseline streams; raw no-floor forecast ≈1,163 streams
WDM 6 keV / FDM 10^{-21} eV	cutoff outside or near edge of sensitive band	effectively indistinguishable in this sample
SIDM	shallower/broader gaps from sufficiently cored perturbers	no abundance handle; morphology AUC depends on core strength

The present seven-stream sample is therefore a null-power regime, not a decisive model-selection regime. The same forecast predicts only E[N_det]=0.18 detectable impacts across seven GD-1-like streams under CDM, so P(0 detections)=0.83 in the floor-normalized baseline and 0.95 without the historical low-rate floor. Favorable suppressed models also predict mostly null samples (P0=0.95-0.96 for WDM 3 keV and FDM 10^-22 eV), giving seven-stream discrimination only Z=0.42-0.45 relative to CDM. Thus the current real-data null is CDM-consistent but not a constraint on WDM/FDM/SIDM; the forecasted population size is the relevant requirement. A 180-case sensitivity grid over rate normalization/floor, completeness, a threshold-completeness proxy, mass band, and stream count leaves this conclusion unchanged: even the most favorable seven-stream cases reach only Z=0.72 for WDM 3 keV and Z=0.79 for FDM 10^-22 eV. The median 3-sigma stream requirements across the grid are 533 and 463 streams, respectively; the threshold axis is a proxy only and should be replaced by measured threshold-specific ROC/completeness curves for any threshold-optimization claim.

8. The timeline forward model

To move beyond detection toward physical characterization, we designed a *timeline forward model* that, for a detected gap, (1) estimates the impact epoch, (2) rewinds the stream to its unperturbed state, (3) re-injects a grid of encounters (mass × epoch × φ_1) with the Erkal et al. 2015 Plummer impulse and the NFW scale radius, (4) re-evolves to the present, and (5) scores each hypothesis against the data. It thereby recasts detection as a constrained, physically explicit fit whose free parameters are the impact's mass, epoch, and location, and it is the natural route around the single-gap degeneracy of §6 (a forward model can exploit the joint density-and-kinematic morphology that a discriminative embedding discards).

Scope caveat (important). The forward model runs on a corrected particle-spray generator with an analytic impulse kick — **not** the validated `streamdf`/`streamgapdf` distribution functions used for detector training. We therefore report its outputs as methods/upper-limit results with a simulator-backend caveat, not as fully DF-ported physical constraints.

Validation and real GD-1. Injecting a known GD-1-like impact (10^9 M⊙, 1.5 Gyr, φ_1=20°) and running the recovery grid returns the truth as the top-ranked candidate (rank 0/36). On the clean STREAMFINDER GD-1 catalog, the model-free finder localizes the known $\varphi_1\approx50°$ gap. The best single-subhalo hypothesis improves only marginally over the smooth null, and after look-elsewhere correction the single-subhalo attribution is not significant: the gap is real, but the present data do not uniquely identify one subhalo as its cause.

Matched-control follow-up diagnostics show that the local real-GD1 fit is sensitive to the assumed smooth-stream background and that tested profiles do not yet improve both density/kinematics and cross-stream morphology together. A matched two-arm `streamdf`/`streamgapdf` backend with continuous perturber scale radius is now implemented and was put through a pre-declared injection/recovery identifiability screen. The screen fails its frozen gates: the recovered profile family is correct only 42% of the time with the real GD-1 selection (58% on idealized, fully-sampled streams) against a 70% gate, and detection recall is 25–33% against a 60% gate. The lowest-mass gaps are too shallow to detect, and detected gaps under-determine the perturber's scale radius (recovery RMSE 0.34–0.41 dex). Selection sparsity worsens but does not cause this; the dominant limit is the intrinsic gap→profile degeneracy. Per the pre-registered rule, real-data profile inference and per-system DM-model typing remain blocked.

> **In plain terms.** The next step beyond "is there a gap?" is "what made it?" We built a tool that rewinds a stream, drops in a simulated clump, fast-forwards, and checks the match. It can recover impacts we plant ourselves and it confirms the GD-1 gap is real, but it cannot yet pin that gap on one specific clump. And when we ask the harder question — what *kind* of clump, read from its density profile (the fingerprint that distinguishes cold from warm or self-interacting dark matter) — even perfect data cannot tell the look-alikes apart, so we make no claim about which dark-matter model made GD-1's gap.

9. Population significance across streams

A single stream rarely yields a decisive detection, so the pipeline includes a multi-stream combination designed to test for a *population* of impacts. Per stream, the best-scoring impact hypothesis is compared against a **data-driven, gap-removed look-elsewhere null**: the null preserves the real stream's broad density envelope and marginal kinematics, removes localized structure, searches the same full grid as the data, and retains its best candidate. Per-stream evidence is then combined with Stouffer's Z and Fisher's method **gated by a coherence requirement**.

Applied to the seven target streams, the correction is decisive: naive single-cell significances as large as 19σ (Sylgr) and 12σ (Pal 5) collapse to about ±2σ after the look-elsewhere correction. The joint result is null — Stouffer Z=1.40 (p=0.08) and Fisher p=0.28, with the coherence gate not satisfied — so the current real-stream result is a

CDM-consistent upper limit, not a dark-matter detection.

> **In plain terms.** A single stream rarely settles the question, so we built a way to pool many streams and to guard against a statistical trap — if you don't account for how many places you looked, pure noise can masquerade as a discovery. Pooling seven streams still gives no convincing signal. That is scientifically useful: it is an upper limit, and it shows how badly naive many-sigma claims can be inflated if the search trials are not counted.

10. Limitations and systematics

1. **Single-gap degeneracy** (§6): mass/geometry/epoch are only partially recoverable from one gap; the forward model and population statistics are the routes around it.
2. **Density-only sensitivity** (§5): weak/old, kinematics-only perturbations are missed. A model-free diagnostic shows the proper-motion kink is real in noise-free simulations but below current Gaia-like precision, so kinematic retraining is a future-data opportunity rather than a current fix.
3. **Action-angle model validity**: clean generation is limited to eccentric streams; near-circular streams (Pal 5, Fjörm) are excluded, and multi-impact streams require `streampepperdf` (unavailable in this galpy), so single-vs- multiple classification is deferred (gap *counting* is available model-free).
4. **Real-data volume**: DM-model discrimination needs many clean detections (§7); today's clean catalogs are few.
5. **Forward model on a separate generator** (§8): the timeline forward model and the multi-stream significance framework run on a corrected particle-spray + impulse backend, not the validated `streamdf`/`streamgapdf` of §3. The quantitative outputs are therefore reported only with this simulator-backend caveat. The single-encounter and impulse approximations also apply. The new matched two-arm DF backend has now been run through a pre-declared injection/recovery identifiability screen and fails its family-recovery (42–58%, gate 70%) and recall (25–33%, gate 60%) gates (§8), so real-data profile inference stays blocked by a measured gap→profile degeneracy.

11. Conclusions

We presented a complete, reproducible pipeline for dark-matter subhalo-impact searches in stellar streams, built on validated stream distribution functions and a pre-flight validation harness. The GINEConv detector reaches test AUC 0.982 with a zero false-positive operating point and is in-distribution on the real STREAMFINDER GD-1 catalog, where it and a model-free gap finder jointly recover the known $\varphi_1 \approx 50°$ gap. With this validated pipeline we (i) mapped detection completeness across impact type — strong for massive, recent impacts and falling for weak, old ones; (ii) showed that single-gap characterization is degeneracy-limited (subhalo mass is essentially unrecoverable from one gap; only recency is weakly constrained); and (iii) reframed DM-model discrimination as a population measurement, quantifying its sample-size cost: favorable WDM/FDM cases can separate from CDM with about two detected impacts, but collecting them requires roughly 300–365 clean streams in the floor-normalized baseline and roughly 1,160–1,370 streams without the low-rate floor; SIDM requires gap-shape information. The corrected timeline forward model recovers planted impacts, but applied to seven real streams it yields no coherent multi-stream detection after look-elsewhere correction (Stouffer $Z=1.40$, Fisher $p=0.28$). The path to a physical measurement is concrete: more clean catalogs, breaking the gap→profile degeneracy that the matched-backend screen shows currently blocks per-system DM typing (via high-precision kinematics and multi-tracer streams), and multi-encounter modeling for crowded streams.

Methods *(technical appendix)*

Potential & integration. galpy `MWPotential2014`; C `dop853` integrator (verified active, no Python fallback); $R_0=8.0$ kpc, $V_0=220$ km s^{-1}. **Distribution functions.** `streamdf` (Bovy 2014) for the smooth track with velocity dispersion σ_v from config; `streamgapdf` (Sanders et al. 2016) with `impactb`, `subhalovel`, `timpact`, `impact_angle`, GM, and r_s. The action-angle setup uses `b = estimateBIsochrone(pot, R/R0, z/R0)` (≈0.61 for GD-1) and `nTrackChunks=5`; `impact_angle` must share the sign of the modeled arm. **Subhalo physics.** NFW scale radius $r_s = r_{200}/c$ with the Ludlow et al. 2016 concentration–mass relation and $\rho_{crit,0} = 277.5\ h^2\ M_\odot\ \mathrm{kpc}^{-3}$; Erkal et al. 2015 Plummer impulse $\Delta v = -(2GM/w)\cdot b/(|b|^2+r_s^2)$. **Detector.** GINEConv encoder, 18 node / 5 edge features, hidden 256, 6 layers, embedding 128, profile branch (48 bins, compact feature set); binary head trained with BCE, auxiliary regression (`mass_time` = [$\log_{10}M$, $\log_{10}t$]) with weight γ_reg; AdamW (lr 3×10^{-4}, wd 10^{-4}); temperature calibration on the validation split. **Noise model.** Gaia DR3-like per-star errors (GaiaDR3); radial velocity dropped at inference (`ignore_rv`) to match clean membership catalogs. Training on contaminated (foreground-injected) streams collapses separability (G6 0.96→0.56), so the operating regime is clean membership catalogs. **Statistics.** Corrected particle-spray/impulse forward-model backend for replay; data-driven gap-removed best-of-grid look-elsewhere null; Stouffer/Fisher combination with a coherence gate; significance via tail probability → z. **Datasets.** `data/simulations_detector_df` (15,830 sims; GD-1, ATLAS, Jhelum, Orphan); detector `checkpoints/detector_df_20260602`.

Reproducibility

Public repository; conda environment spec; 330 passing unit tests and 9 skipped checks; categorized `scripts/` index; per-decision changelogs. Datasets regenerate from `scripts/generate_detector_data.py` (deterministic per seed, resumable); figures from `paper/make_figures.py`, `paper/make_report_figures.py`, and `paper/make_report_figures2.py`; validation from `scripts/validate_generator.py`; claim provenance in `paper/claims_audit.md`.

References

Works cited (full entries in `references.bib`): Banik et al. (2021); Bonaca et al. (2019); Bovy (2014, `streamdf`); Bovy (2015, `galpy`); Carlberg (2012); Erkal & Belokurov (2015); Fardal et al. (2015, `streamspraydf`); Gaia Collaboration (2023, DR3); Hu et al. (2020, GINEConv); Ibata et al. (2021, STREAMFINDER); Ludlow et al. (2016); Mateu (2023, `galstreams`); Price-Whelan & Bonaca (2018); Sanders, Bovy & Erkal (2016, `streamgapdf`).